Boise State University

Department of Electrical and Computer Engineering

ECE 350L Circuit Analysis and Design Lab

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Lab Report

Experiment #11 AM Modulation

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1. Introduction

This report is an investigation of Amplitude Modulation (AM) where it is used to send information through signals. In this lab, AM signals are created where they are then modulated together. The modulated signals allow them to be transmitted though multiplexing to be sent over long distances. Later on in the report, the multiplexed signal is demultiplexed and demodulated to separate individual signals. The signals created in the beginning of this report, should be maintained at the end of the report to ensure the process is correct.

The objective of this report is to explore the concept of Amplitude modulation and filtering through practical implementation. This is done by utilizing a software MATLAB by creating signals then plotting the varying steps. The Amplitude modulation will prove correct if the original signals are recovered from the modulation and multiplexing processes.

Throughout the report, connections between time and frequency domain are referenced. The processes of modulation, multiplexing, filtering, and wave rectifiers demonstrate how signals with information are sent throughout the world in communication systems. Communication systems allow information such as text messaging and calls to be transmitted to other parts of the world.

2. Theory

Amplitude Modulation is a type of signal modulation where the amplitude of a waveform is modulated. Modulation is the process of altering a signal such as rectangular waveform. Amplitude Modulation is commonly used in communication systems for example, broadcasting over radio frequencies. When a signal or waveform is modulated, demodulation is needed to remove the carrier frequency. The purpose of modulation is to efficiently transmit signals over distances. If there are multiple signals that needed to be transmitted, overlapping would occur where the receiver could not understand without decoding.

The general process of this experiment is drawn as a block diagram found in figure 1. The block diagram steps through each process of modulation, multiplexing, demultiplexing, and passing through a set of filters to recover the original signal. The general process in figure 1 is present in the real world however, more complex to achieve a specific outcome.

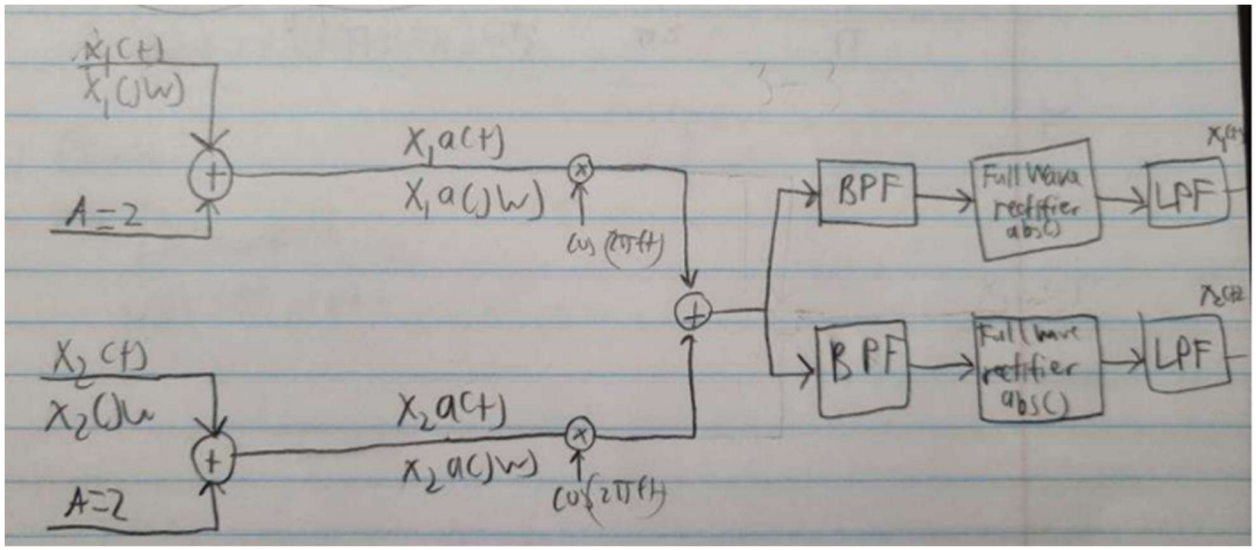

Figure 1: General Block Diagram of report

3. Procedure and Results

3.1 Creating Signals to Modulate

Two signals are created named x1(t) and x2(t) utilizing piece wise functions in MATLAB. Signals x1(t) and x2(t) are simplification of signals sent by the ordinary person such as text messaging. The signals x1(t) and x2(t) represent different communication patterns shown in figure 2.

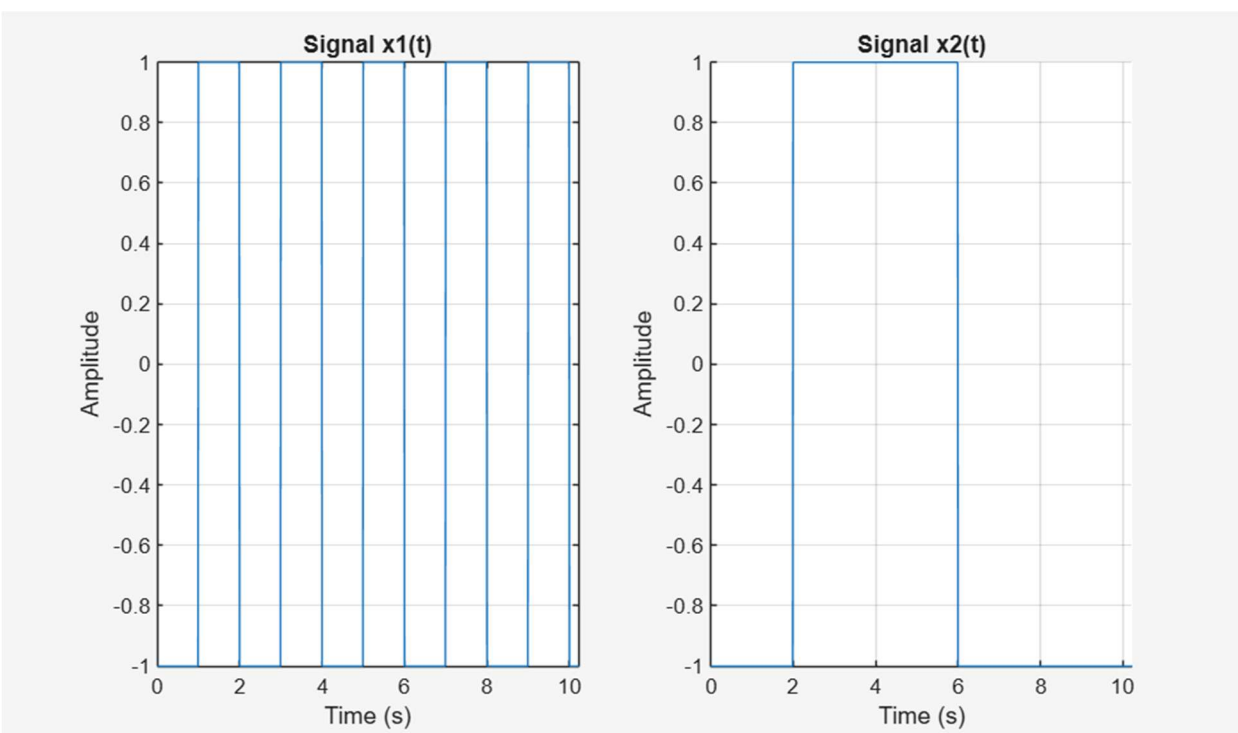

Figure 2: Base Signals (2a) x1(t) and (2b) x2(t)

The bit pattern of Signal x1(t) is 0-1-0-1-0-1-0-1-0-1-0 where x2(t) is 0-1-1-1-1-0-0-0-0. Each signal has their own unique waveform to differentiate between. The unique waveforms x1(t) and x2(t) act as information say in a text message. Signals x1(t) and x2(t) will be used to show the separate signals and determine if the output is maintained.

3.2 AM Modulating Signals

To prepare both signals for transmitting, an offset amplitude will be applied to each signal individually. The purpose of adding an offset of 2 before modulation is when the cosine carrier is multiplied to each signal, the signal would become negative. Meaning the envelop would cross as zero. Figure 2 shows the original signals x1(t) and x2(t) both with an offset of 2 added.

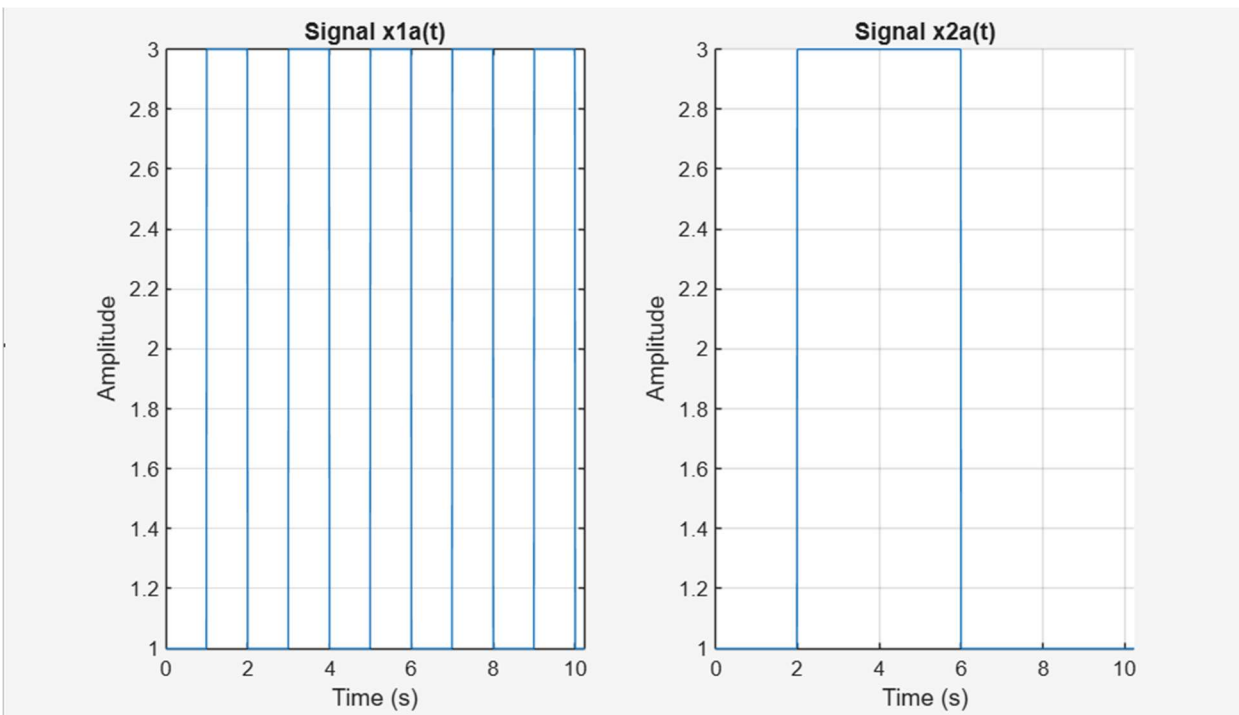

Figure 3: Signals (3a) x1(t)+2 offset and (3b) x2(t)+2 offset

Comparing the signals in figure 2 to figure 2, both maintain their original rectangular shapes. However, the offset of 2 applied to when the signals on, are now amplitude of 3 instead of 1. Next, the signal will be multiplied by a cosine carrier of 15 Hz and 25 Hz to x1a(t) and x2(t) respectively. Figure 4 shows the amplitude signals with modulation with a cosine carrier.

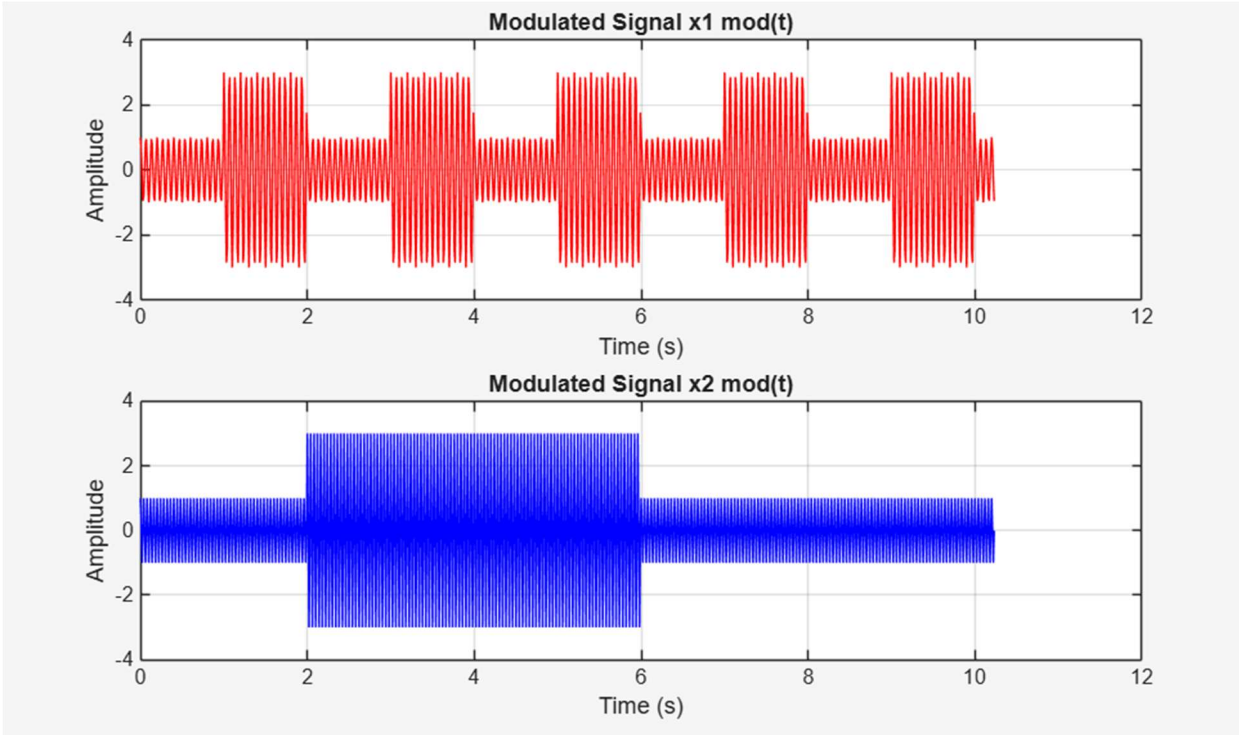

Figure 4: (4a) Signals x1a(t) modulated with 15 Hz and (4b) x2a(t) modulated with 25 Hz

Performing the modulation with cosine carriers allows for the amplitude modulation. The modulation allows for the signals x1a(t) and x2a(t) to shift each spectra into clear frequency bands. In figure 4, the center frequency are 15 Hz and 25 Hz where in figure 4a, the lower frequency has lower oscillations compared to figure 4b. This is respective to the 15 Hz of x1(t) and 25 Hz of x2(t).

Next a Fast Fourier Transform (FFT) can be generated to provide an interpretation in the frequency domain. A FFT in this report is an algorithm used to convert time domain signals to the frequency domain. The modulation process shows the shift of the base signals (x1(t) and x2(t)) to the new spectra where the center frequencies are 15 and 25 Hz. Figure 5 illustrates the FFT in the frequency domain.

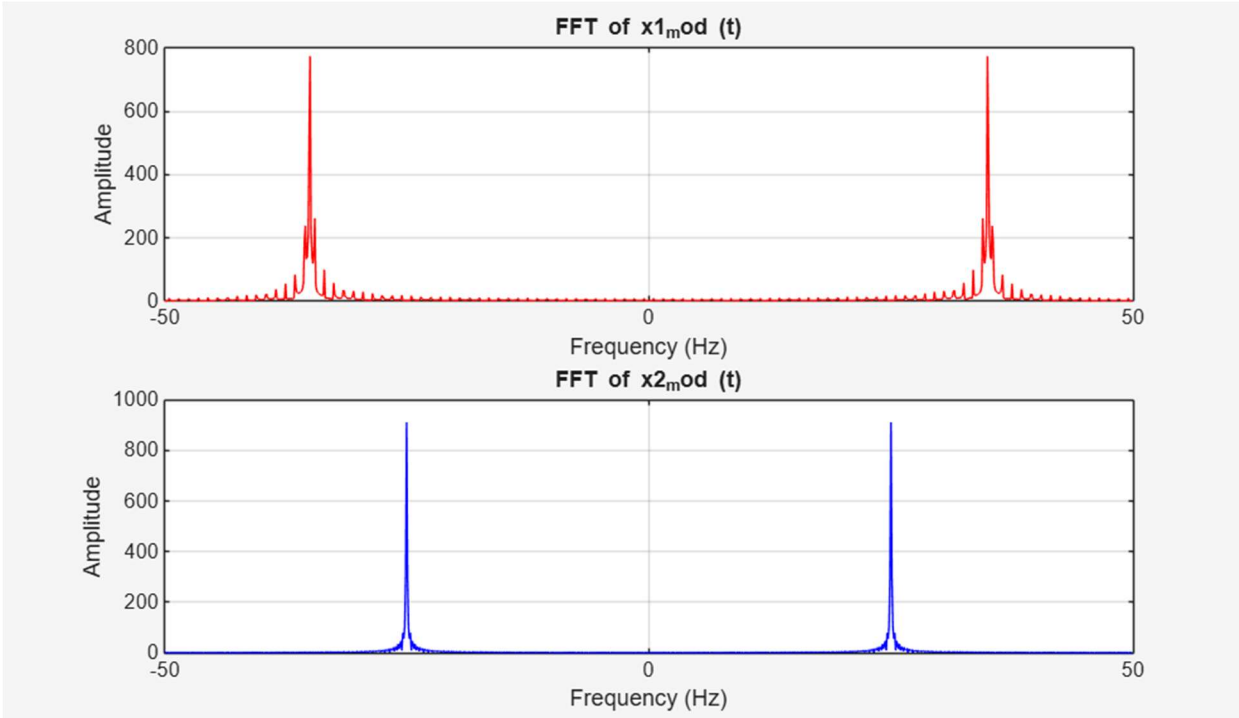

Figure 5: (5a) FFT plot of x1mod(t) and (5b) x2mod(t)

The FFT of x1mod(t) and x2mod(t) in the frequency domain clearly show their peaks at their respective frequencies of 15 Hz and 25 Hz. Each signal has their own frequency regions allows them to not overlap before multiplexing. The signals in their own frequency prevents aliasing from occurring which can distort say the information carried in the signal.

3.3 Multiplexing AM Signals

To further streamline the transmission, combining both signals of x1mod(t) and x2mod(t) allows for a single signal to be sent. The combined signal allows for easier transmission and reduces the amount of signals need to be sent and retrieved. Thus, a FFT will be used to create a combine signal.

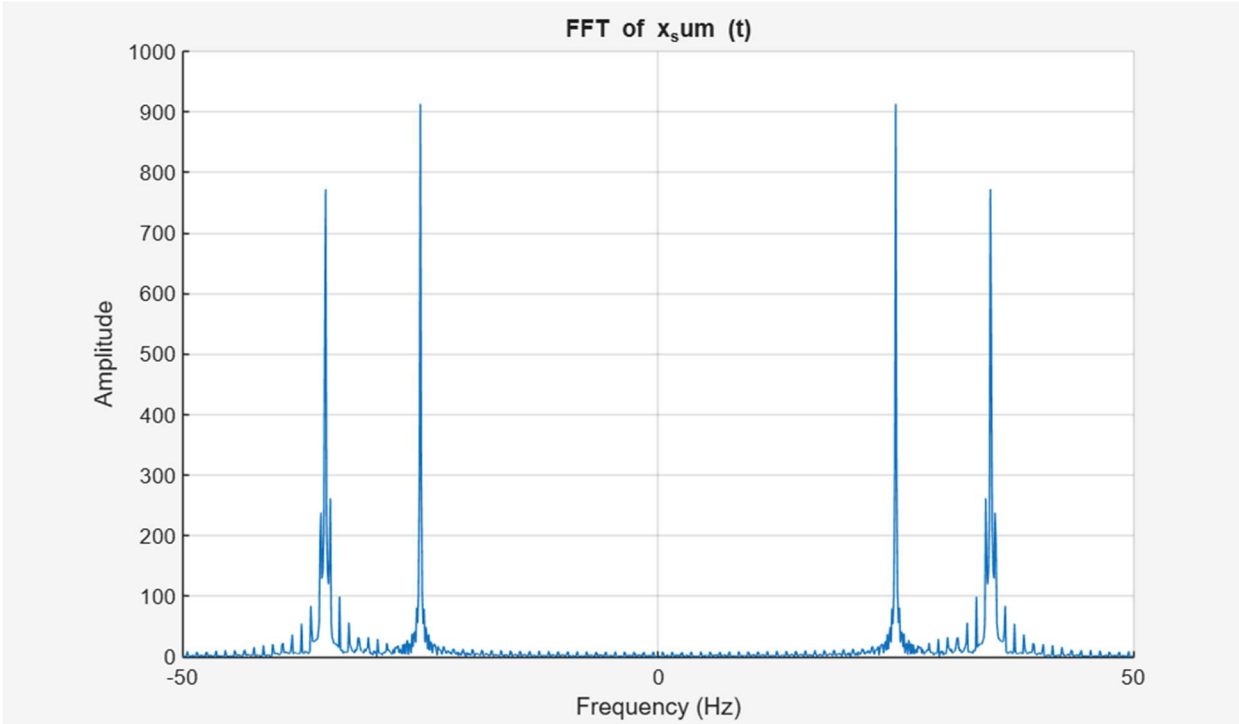

Figure 6: FFT of xamod(t) and xbmod(t) multiplexed

The multiplexed signals of signals x1 and x2 modulated shows the maintain frequency's at 15 Hz and 25 Hz. Both signals maintained their individual upper and lower sidebands as well as their peaks. In this stage, the multiplex signal would be transmitted to a receiver where it would then be demultiplexed to their original unique signal.

3.4 Demultiplexing AM Signals

In order to separate the signals from figure 6, demultiplexing is utilized through two separate band pass filters. The bandpass filters isolate the center frequencies of 15 Hz and 25 Hz. Isolating the center frequencies attenuate unwanted spectra while maintaining the original signal. Each bandpass filter of 15 Hz and 25 Hz is show in figure 7

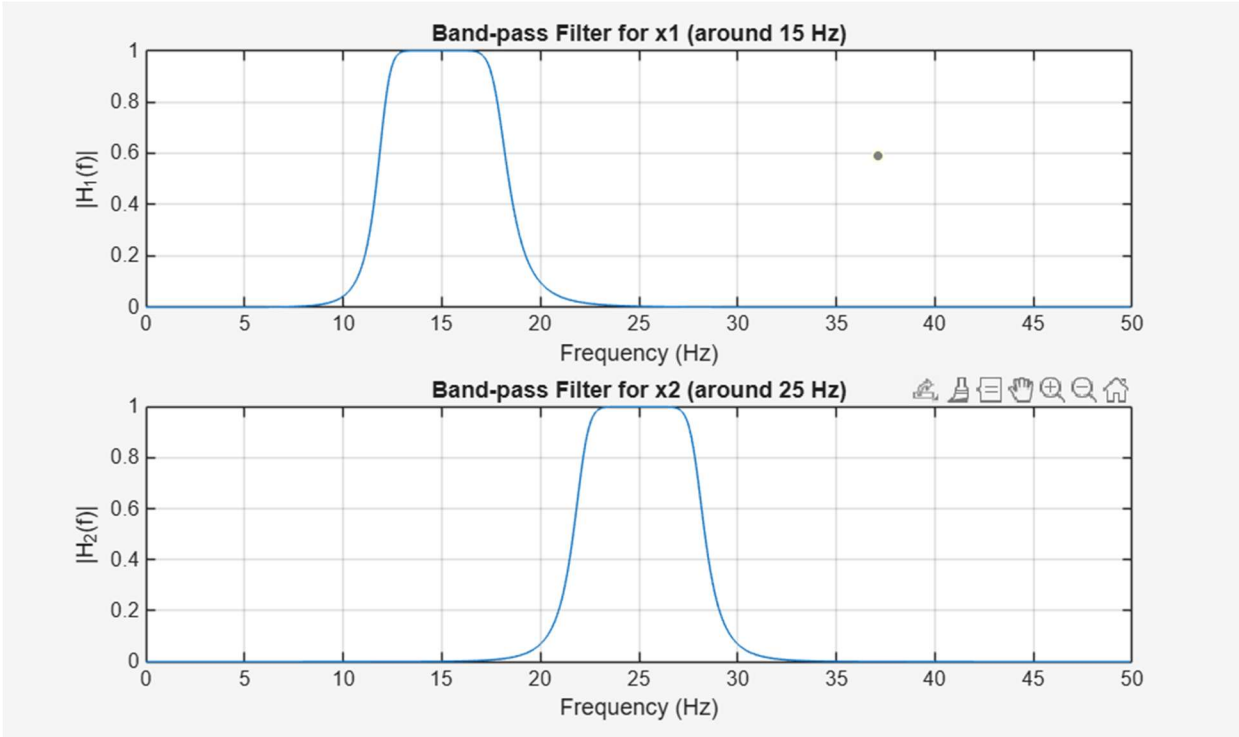

Figure 7: Bandpass filters at (7a) 15 Hz and (7b) 25 Hz

The bandpass filters in figure 7a and 7b are used to extract the signals at 15 Hz and 25 Hz respectively from figure 6. The signal from figure 6 is passed through each bandpass filter where in the 15 Hz filter will attenuate signals not equal to 15 Hz. For example, a signal at 25 Hz will be filtered out in the 15 Hz bandpass filter. Thus, the separated signals are shown in figure 8a and 8b in the frequency domain.

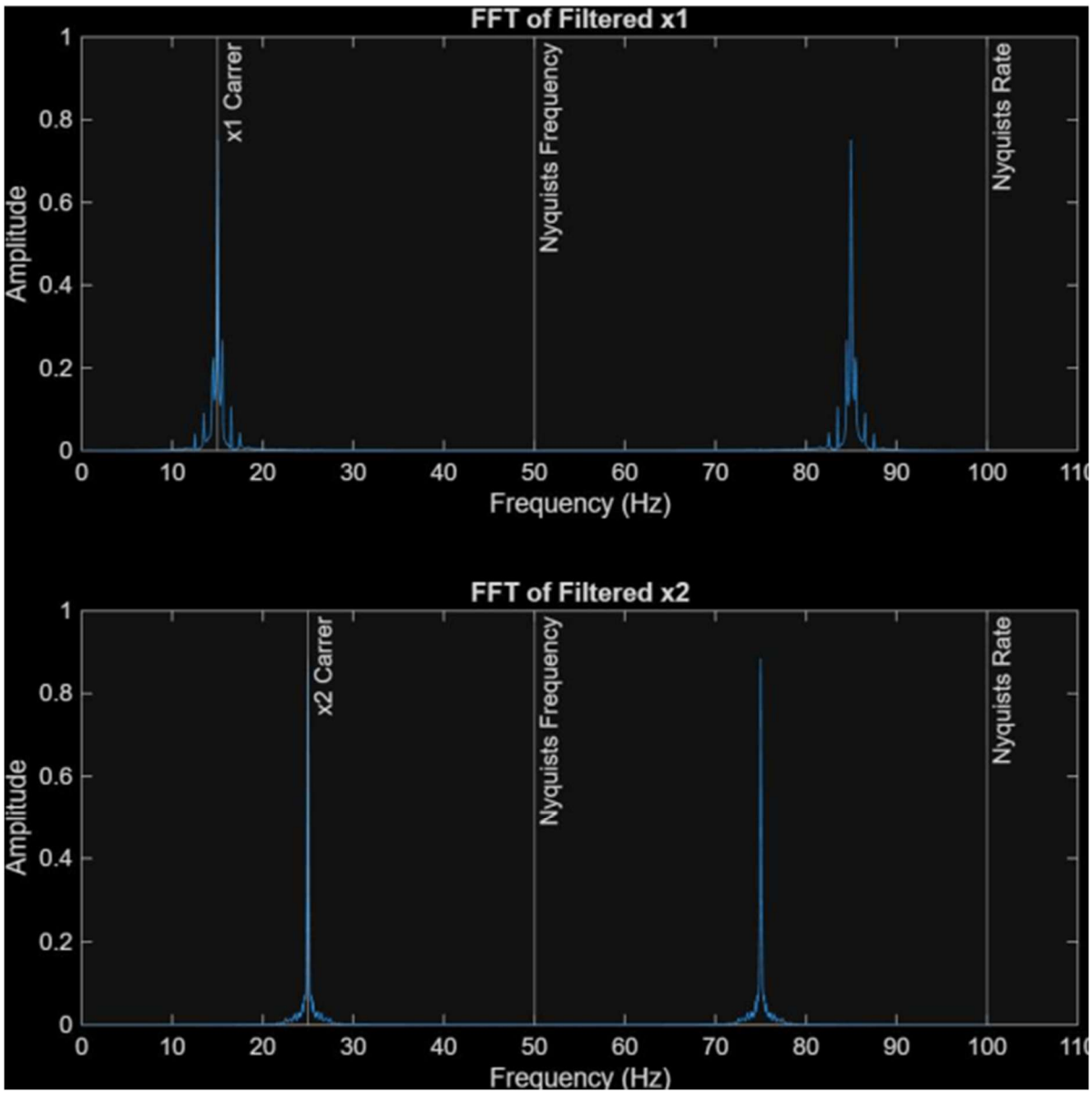

Figure 8: FFT filtered signal at (8a) 15 Hz and at (8b) 25 Hz

Comparing outputs of figure 8 and figure 5, the signals match proving the bandpass filters worked. Nyquist frequency is displayed as 50 Hz and Nyquist rate is 100 Hz. The Nyquist frequency of 50 Hz proves the rate was 100 Hz. The filtering confirms true as frequencies of 15 Hz and 25 Hz are present without overlapping. The signals are also plotted in the time domain to further analyze shown in figure 9.

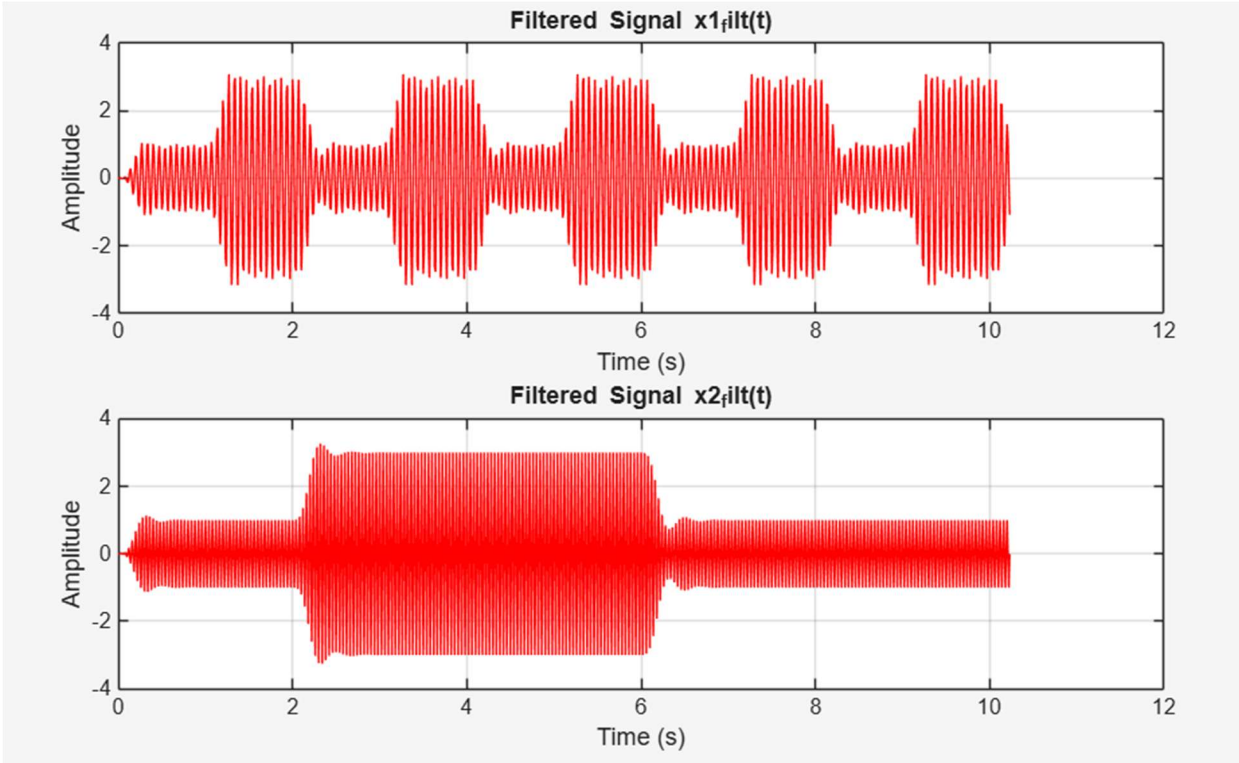

Figure 9: (9a) filtered signal x1(t) (9b) filtered signal x2(t)

The signals are successfully retrieved after passing through band pass filers. In comparison to figure 4, the signals are similar. The general shape of the signals from figure 9 resemble figure 4 signal shapes. Figure 9 signal shapes are smoother in comparison to figure 4 due to bandpass filters. In an ideal setting a bandpass filter would only extract frequencies of 15 Hz and 25 Hz. In practice, a range of frequencies can only be extracted as the information in the signal is maintained. The separation of signals are now able to be used to demodulate and produce a final outcome shown in figure 10.

3.5 Demodulating AM Signals

In order to demodulate the AM signal, the cosine signals must be removed from each x1(t) and x2(t). Two methods are used in practice which are synchronous and asynchronous demodulation. Asynchronous demodulation will be used as it's ease of implementation. In asynchronous demodulation, a full-wave rectifier will be used for each signal. The purpose of a full-wave rectifier is to pass all values as positive integers. The positive values would remain unchanged where the negative sign values are turned to positive sign. Figure 10 shows the

signals x1(t) and x2(t) passing through full-wave rectifiers. [1]

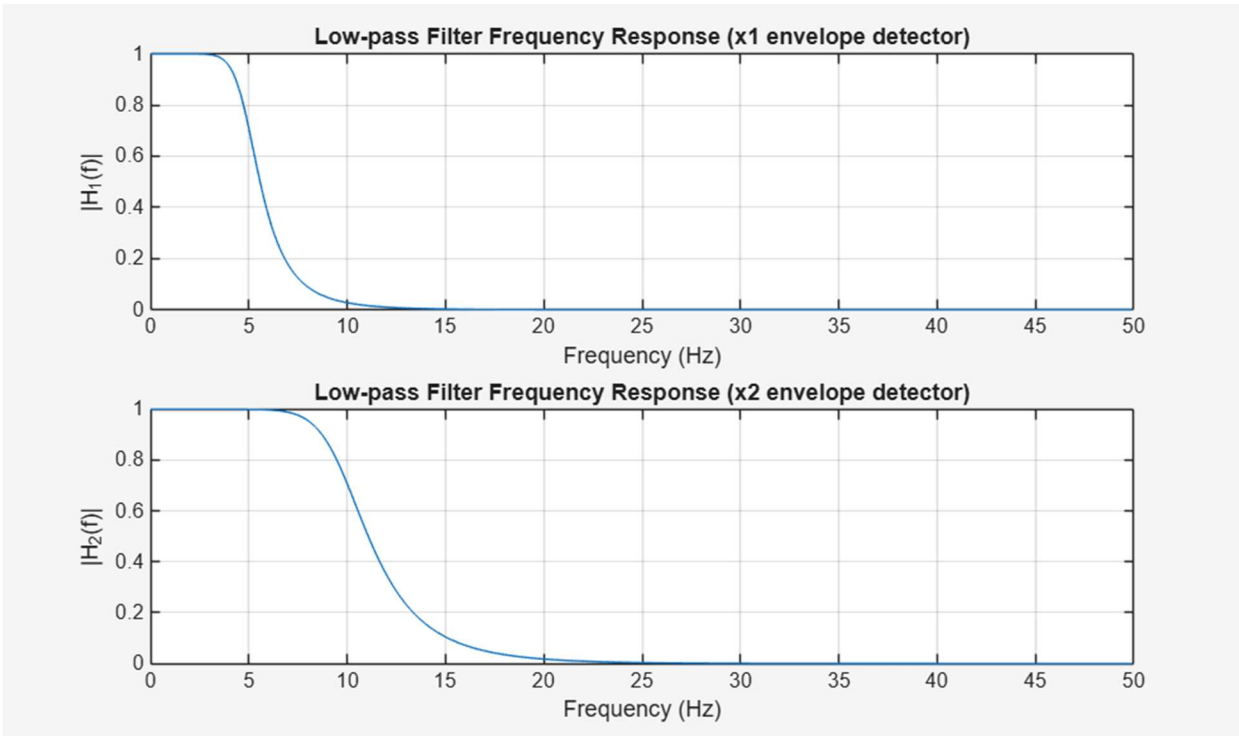

Figure 10: (10a) LPF frequency response of x1, (10b) LPF frequency response of x2

The plot in figure 10a and 10b show how each respective signal are decreasing until they reach an amplitude of 0. As the signals from figure 10a and 10b are passed through the full-wave rectifier, other frequencies outside of 15 Hz and 25 Hz are attenuated. The result is cosine carrier removed from each signal, leaving behind a smoother signal. Each signals x1 and x2 are sent through a final low pass filter shown in figure 11.

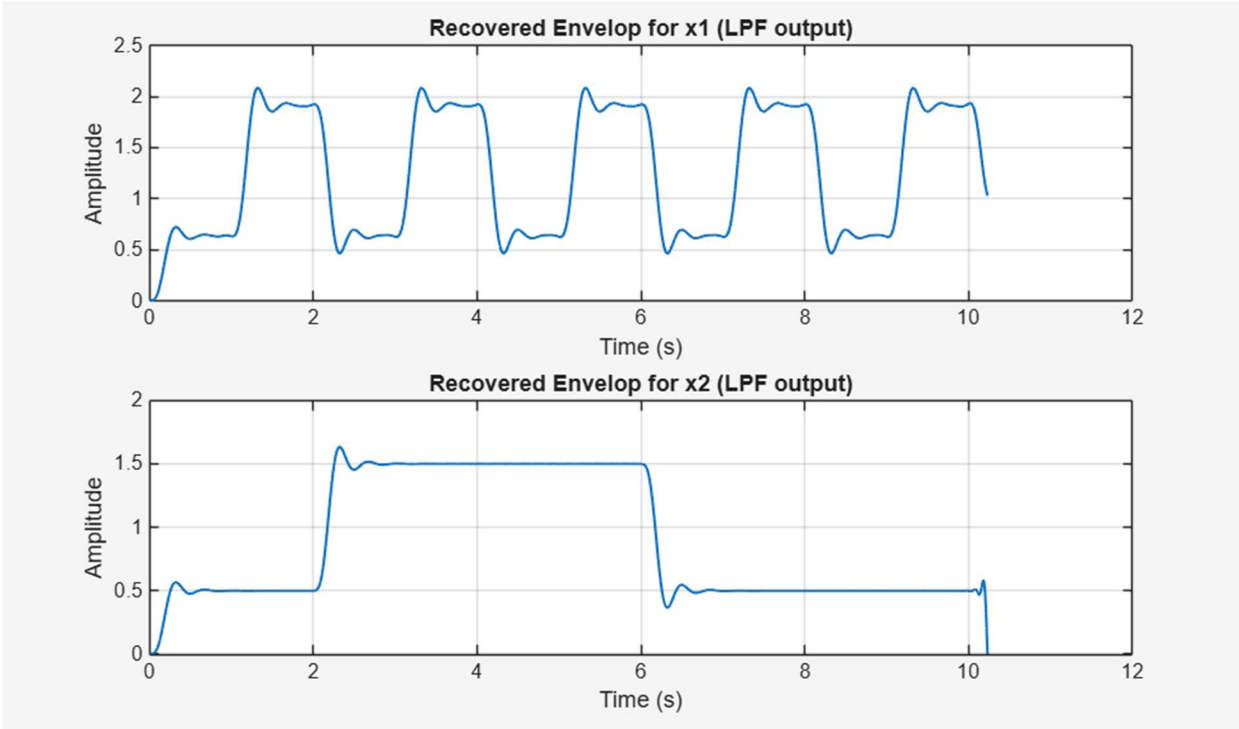

Figure 11: (11a) recovered x1 signal, (11b) recovered x2 signal

The low pass filter attenuates higher frequencies where the recovered signals are shown in figure 11a and 11b. Comparing figure 11 to figure 2, the rectangular shapes are recovered. However, the amplitudes do not match the input signals because of the mutliple processes such as AM modulation, multiplexing, full-wave rectification, and filtering. The process distorts the signals x1 and x2 to behave differently each time. As a result, the signal is not perfectly recovered. The spikes seen in figure 11a and 11b when the signal turns high and low are a result of the bandpass and full-wave rectifiers. The lowpass filters removes majority of the unwanted noise but small remainers are still present. Overall, the signals x1 and x2 general shape and timing are recovered from the original waveforms. The amplitudes and exact shape are not exact compared to figure 2 because of the asynchrounous demodulation. Further methods are necessary to fully recover the exact waveform found in figure 2.

5. CONCLUSION

The experiment was successful in sending information like signals through AM modulation. The process was achieved through creating two unique signals where they are

modulated with a cosine carrier, then multiplexed together. The single combine signal was then passed through two separate frequency bandpass filters to extract the specific signals. Later passed through full wave rectifiers then a low pass filter to recover the shape and timing of the signals. The overall process is a simplification of how more complex signals in communication for example are transmitted.

The results achieved through AM Modulation presented the final output signals showing resemblance of the original signals. While the signal is not an exact match of the input signals, the theory AM Modulation was achieved conceptually through simulations ran in MATLAB. The experiment proved how signals can be condense and separated later on to recover the original signal. As well as showing how further processes is needed to completely restore signals. The processes are a fraction of the how complex systems are in the real world.

REFERENCES

[1] Boise State University, Dept. of Electrical and Computer Engineering, “ECE 305L: Signals and Systems Lab Lab 11 Amplitude Modulation and Demodulation Assignment” 2025.

[2] “What is Amplitude Modulation? — Keysight Oscilloscope Glossary - Keysight Technologies,” *Keysight.com*, 2022. https://www.keysight.com/used/us/en/knowledge/glossary/oscilloscopes/what-is-amplitude-modulation